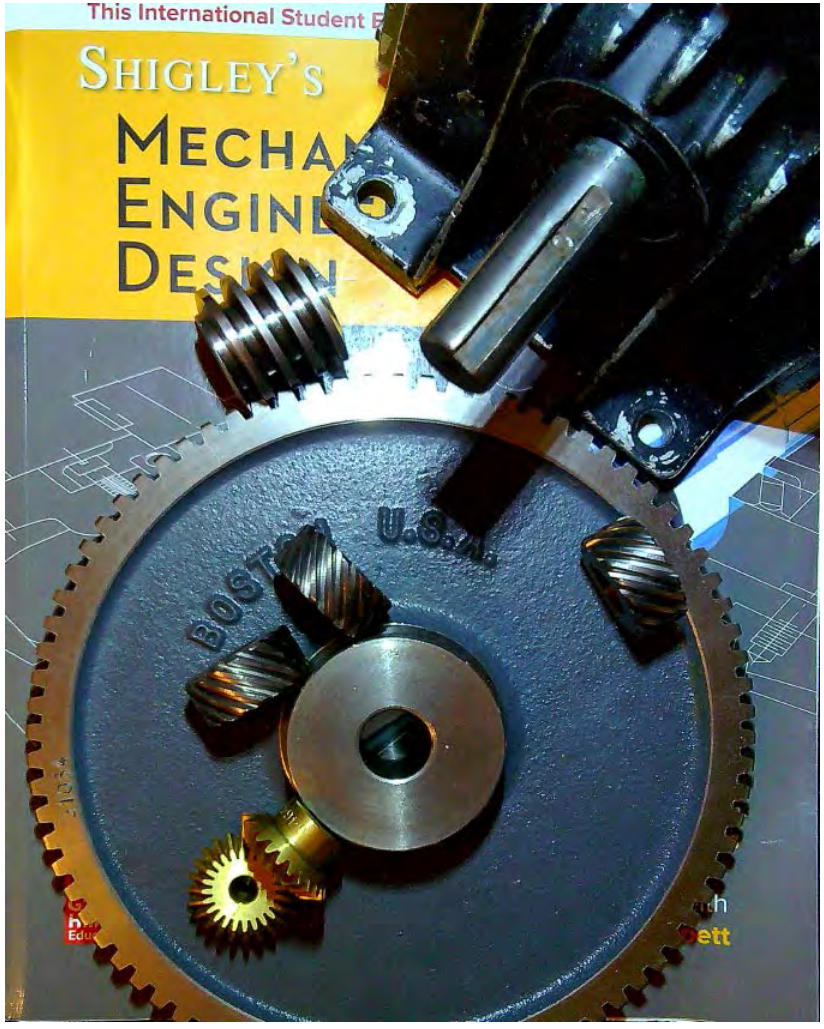
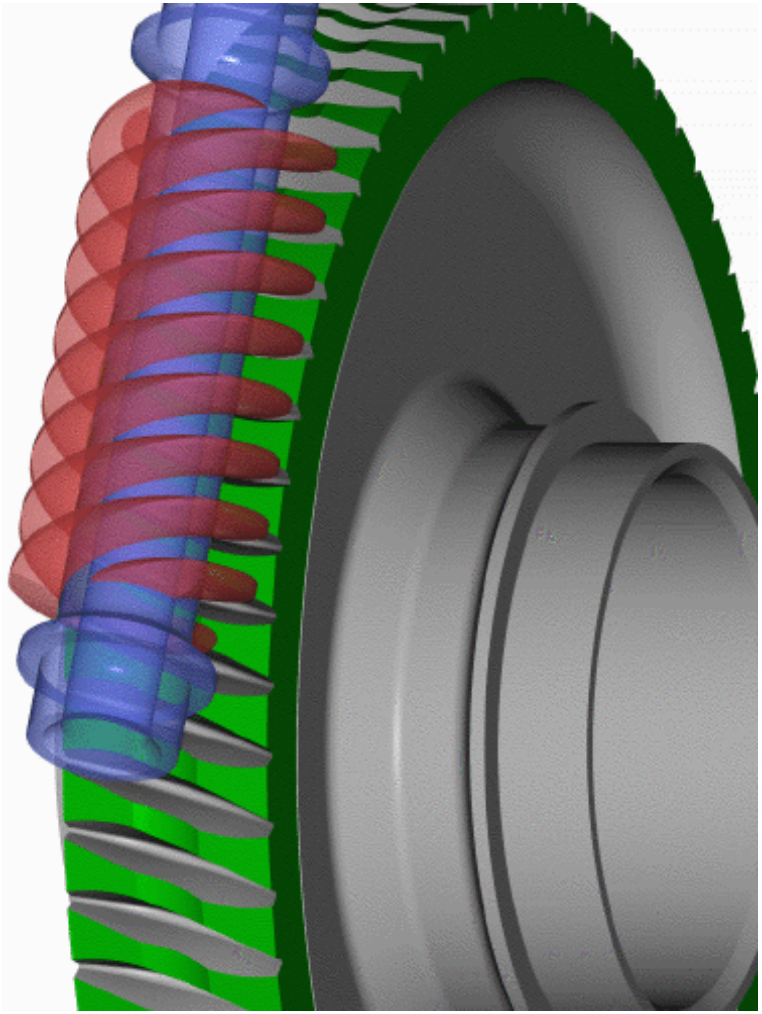
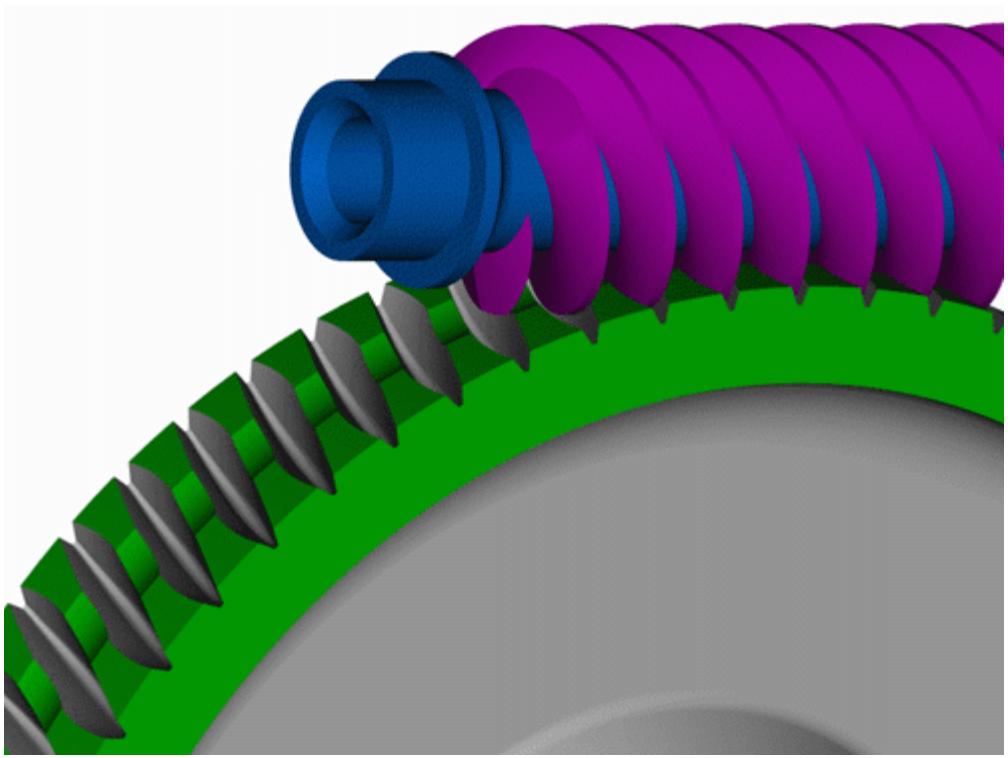
MECH 325
Worm Gears



Worm Gear Animation



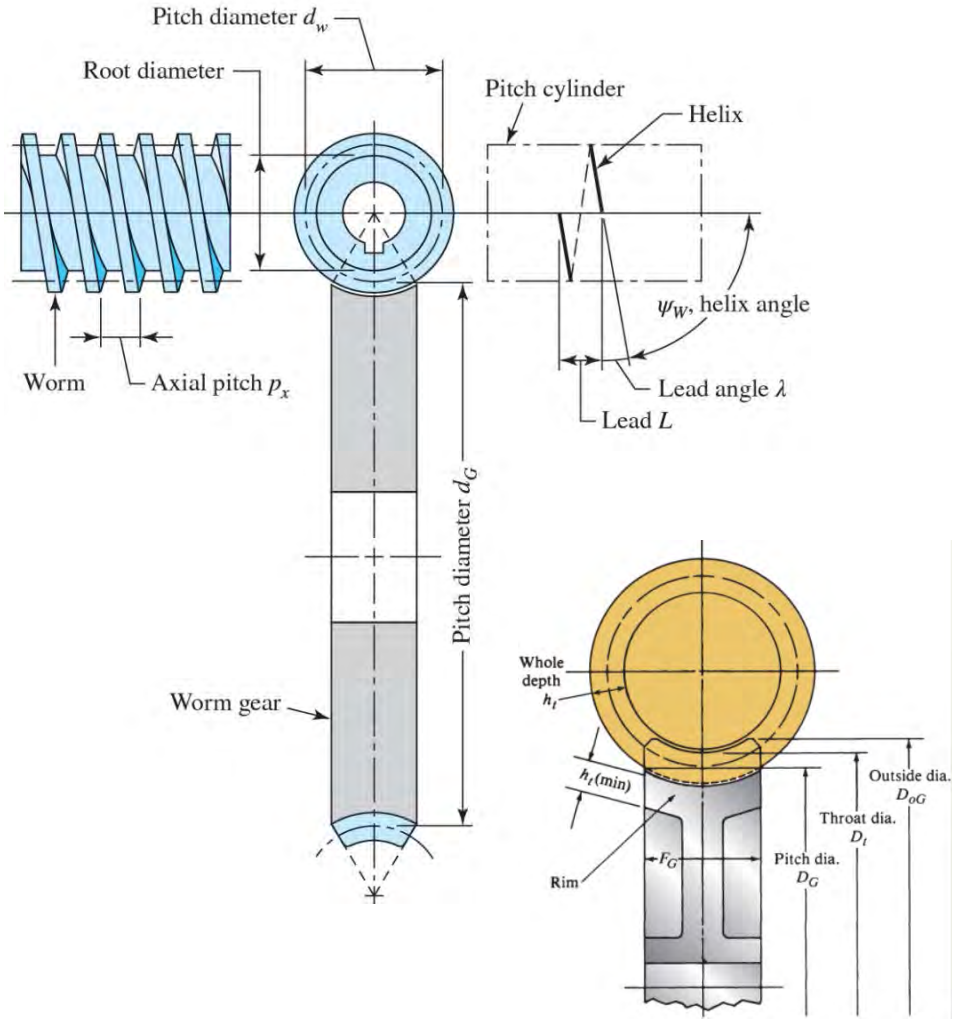


http://www.gearideas.com/tech/warm.htm

Worm Gear

- Common to specify lead angle λ for worm and helix angle ψ_G for gear.
- Common to specify *axial pitch* p_x for worm and *transverse circular pitch* p_t for gear.
- Pitch diameter of gear is measured on plane containing worm axis.

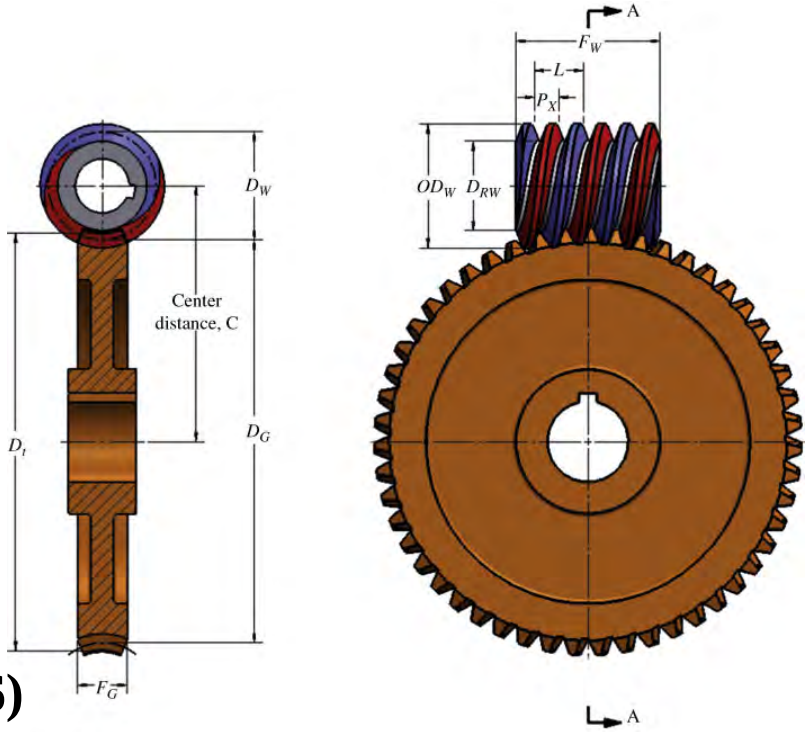
$$d_G = \frac{N_G p_t}{\pi}$$



Worm Gear

- Worm may have any pitch diameter.
- Worm can be cut to be single, double or triple threads
- Should be same as hob used to cut the gear teeth.
- Recommended range for worm pitch diameter as a function of center distance *C*,

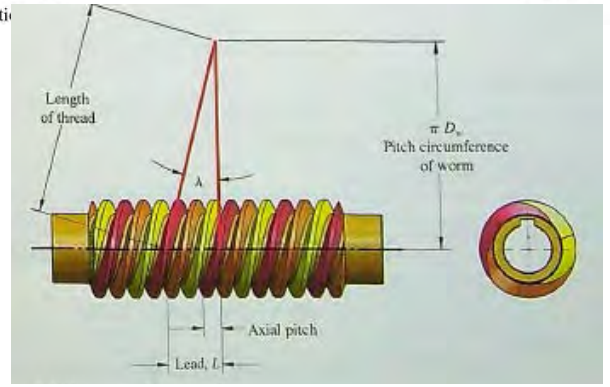
$$\frac{C^{0.875}}{3.0} \le d_W \le \frac{C^{0.875}}{1.7} \qquad \textbf{(13 - 26)}$$



- Relation between *lead L* and *lead angle λ*,

$$L = p_x N_W \qquad \textbf{(13 - 27)}$$

$$\tan\lambda = \frac{L}{\pi d_W} \qquad \textbf{(13 - 28)}$$

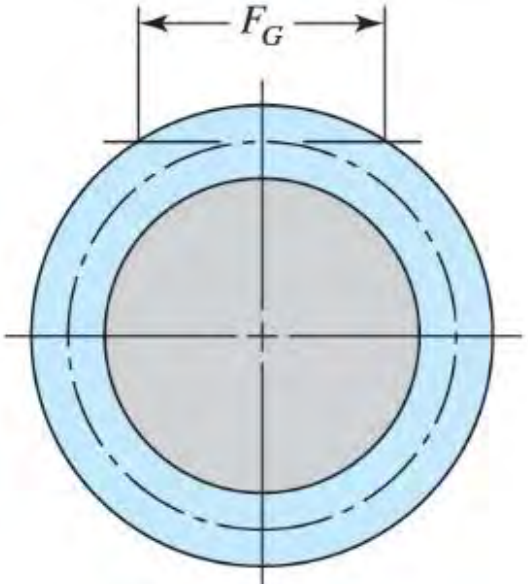


Worm Gear

- The velocity ratio

$$VR = \frac{speed\ of\ worm}{speed\ of\ gear}$$

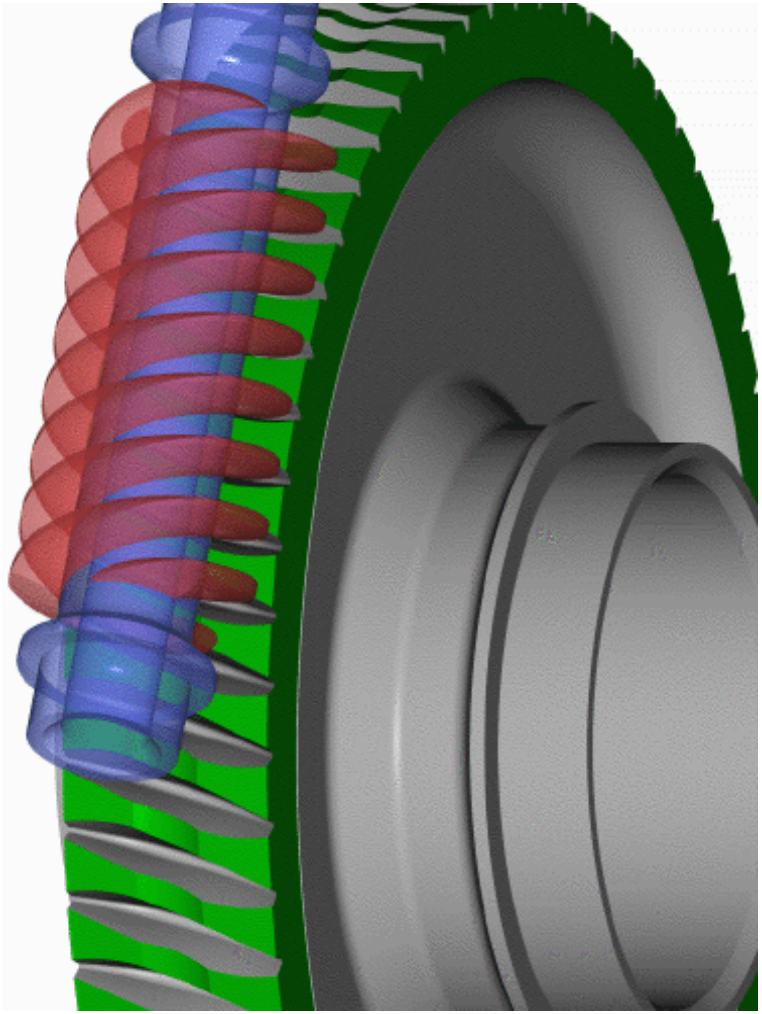
- *Face width* F_G of a worm gear should be equal to the length of a tangent to the worm pitch circle between its points of intersection with the addendum circle.

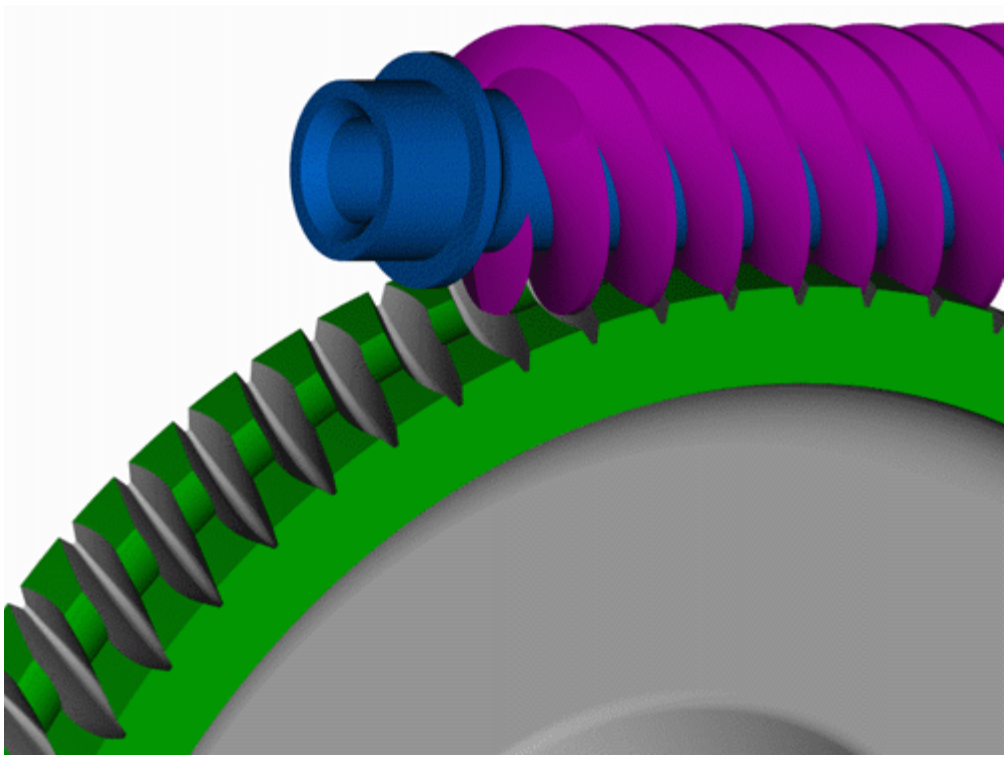


ϕ_n	Maximum Lead Angle λ_{max}
14.5°	16°
20°	25°
25°	35°
30°	45°

Table 15–9 Largest Lead Angle Associated with a Normal Pressure Angle ϕ_n for Worm Gearing

Worm Gear Animation





http://www.gearideas.com/tech/warm.htm

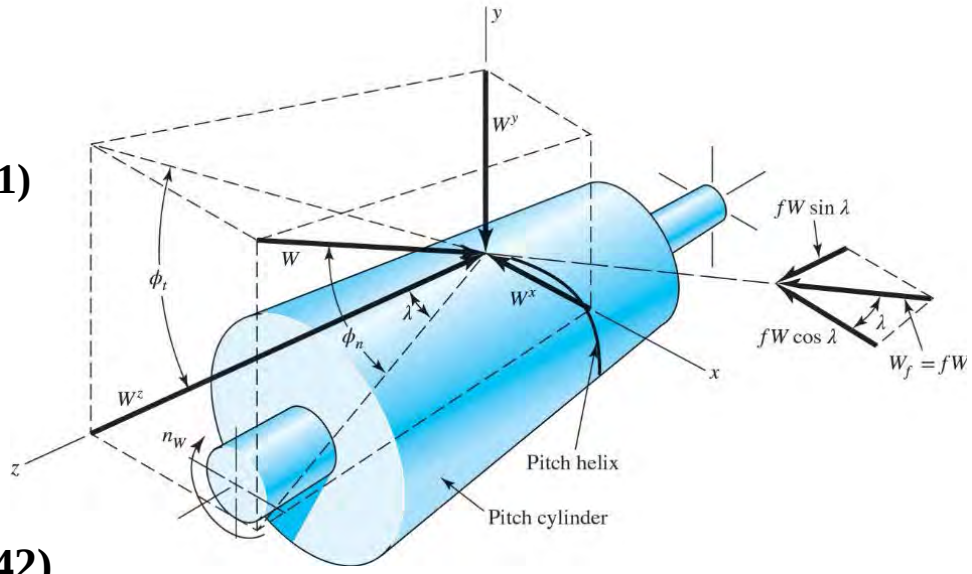
Worm Gear Forces

- Recall forces for worm gears:

$$W^x = W\cos\phi_n \sin\lambda$$
$$W^y = W\sin\phi_n \qquad \textbf{(13-41)}$$
$$W^z = W\cos\phi_n \cos\lambda$$

Forces on the worm translate to the gear forces.

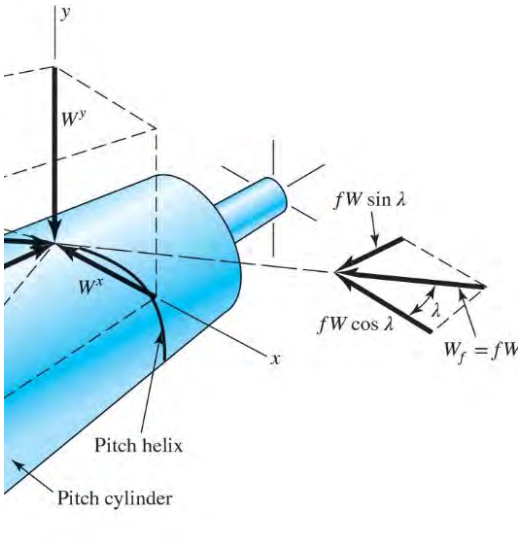
$$W_{Wt} = -W_{Ga} = W^x$$
$$W_{Wr} = -W_{Gr} = W^y \qquad \textbf{(13-42)}$$
$$W_{Wa} = -W_{Gt} = W^z$$



Worm Gear Forces

- Relative motion in worm gearing is sliding action.
- Friction is much more significant than in other types of gears.
- Including friction components, Eq. (13–41) can be expanded to

$$W^x = W(\cos\phi_n \sin\lambda + f\cos\lambda)$$
$$W^y = W\sin\phi_n \qquad \textbf{(13 - 43)}$$
$$W^z = W(\cos\phi_n \cos\lambda - f\sin\lambda)$$



- Combining with Eqs. (13–42) and (13–43),

$$W_f = f\,W = \frac{f\,W_{Gt}}{f\sin\lambda - \cos\phi_n\cos\lambda} \qquad \textbf{(13 - 44)}$$

$$W_{Wt} = W_{Gt}\frac{\cos\phi_n\sin\lambda + f\cos\lambda}{f\sin\lambda - \cos\phi_n\cos\lambda} \qquad \textbf{(13 - 45)}$$

$$W_f = \frac{fW^t}{\cos\lambda\cos\phi_n} \qquad \textbf{(15 - 29)}$$

Friction Coefficient *f*

- Need to consider both static friction and dynamic friction coefficient.
- *Efficiency η (e)* is defined as

$$\eta = \frac{W_{Wt}\,(\text{without friction})}{W_{Wt}\,(\text{with friction})} \qquad (\boldsymbol{a})$$

- Set ratio of W_{Wt} / W_{Wt} with f = 0 in the numerator (13-45),

$$\eta = \frac{\cos\phi_n - f\tan\lambda}{\cos\phi_n + f\cot\lambda} \qquad \textbf{(13-46)}$$

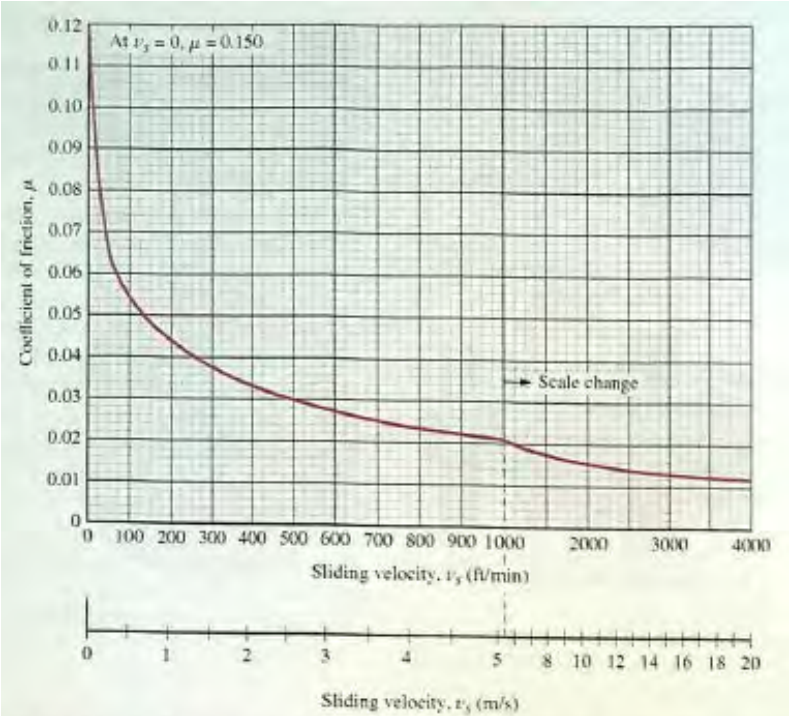
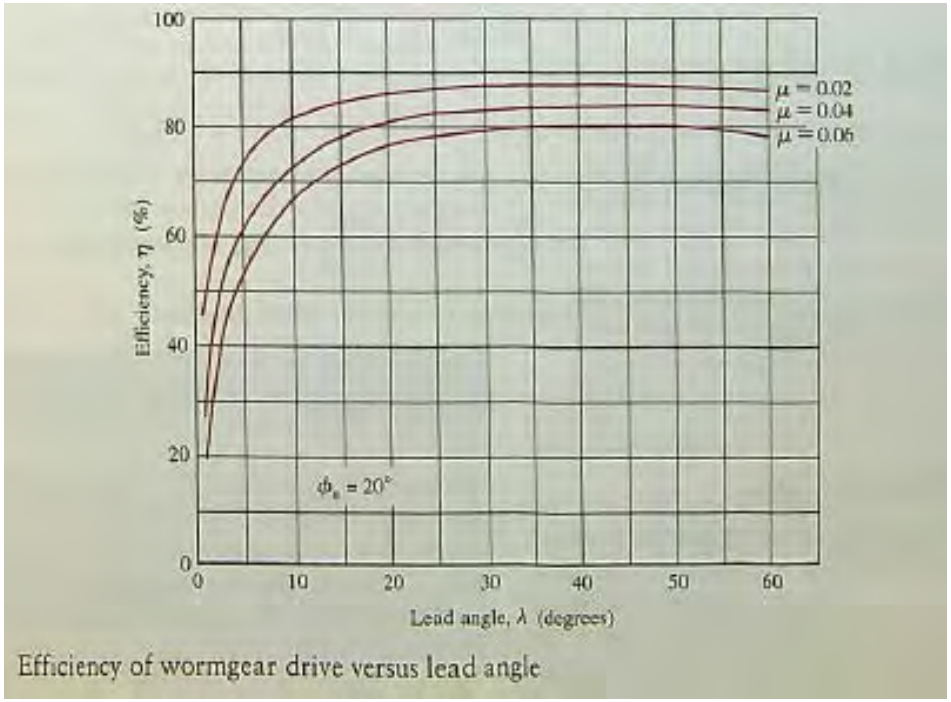


FIGURE 10–25 Coefficient of friction versus sliding velocity for steel worm and bronze wormgear

Worm Gear Efficiency

- Efficiency is reduced significantly at Lead Angles, $\lambda < 20$
- The coefficient of friction must be kept low through material selection and lubrication
- Power losses are product of sliding velocity and friction forces



Efficiency of wormgear drive versus lead angle

$$P_i = P_o + P_L$$

$$P_L = \frac{v_s W_f}{33\,000}$$

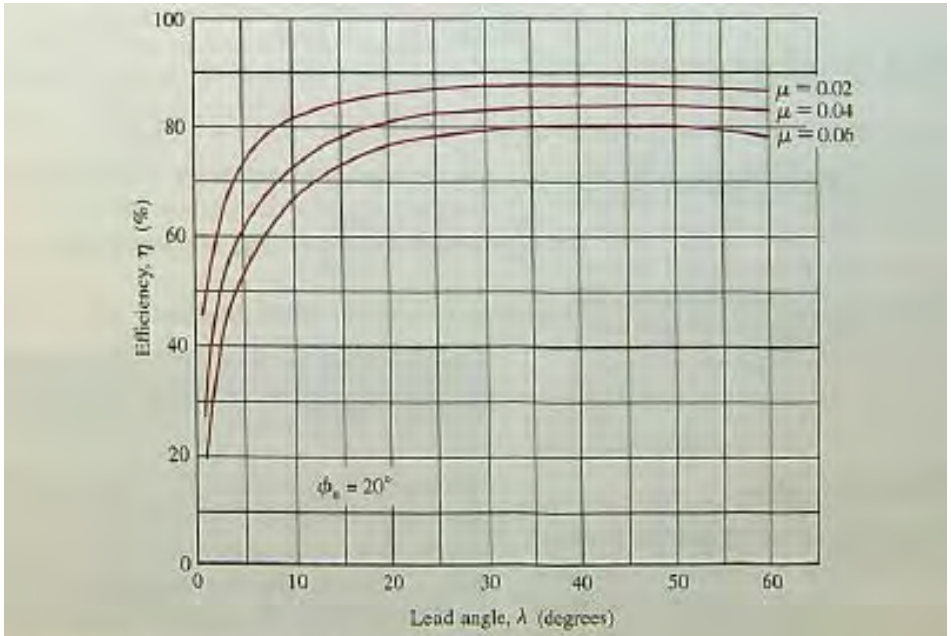
W_f Sliding Force
v_f Sliding Velocity

$$\eta = P_o / P_i$$

Self-locking wormgear sets

- Self-locking worm gears ensure that loads on the gear do not turn the worm. This is normally referred to as *back-driving*
- Self-locking is primarily a function of lead angle. In general, $\lambda < 6°$ ensures self-locking.
- The static friction of worm gears is approx. 0.15 and be compared with the pressure and lead angle.

$$f_{static} > \cos \phi_n \tan \lambda$$



Efficiency of wormgear drive versus lead angle

Worm Gear Velocity

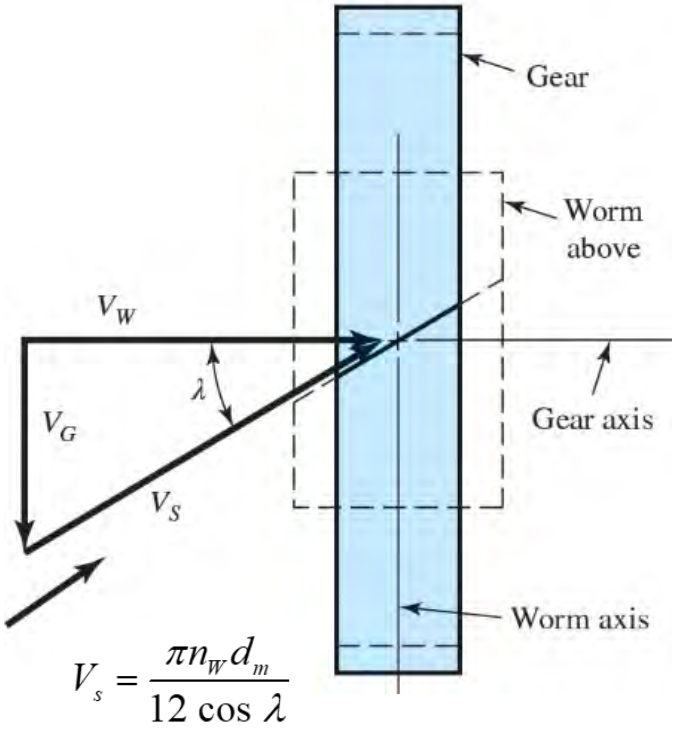
- Coefficient of friction is dependent on sliding velocity V_S.

- V_W is pitch line velocity of worm.

$$V_W = V_S cos\lambda = \frac{\pi d_w n_w}{12}$$

- V_G is pitch line velocity of gear.

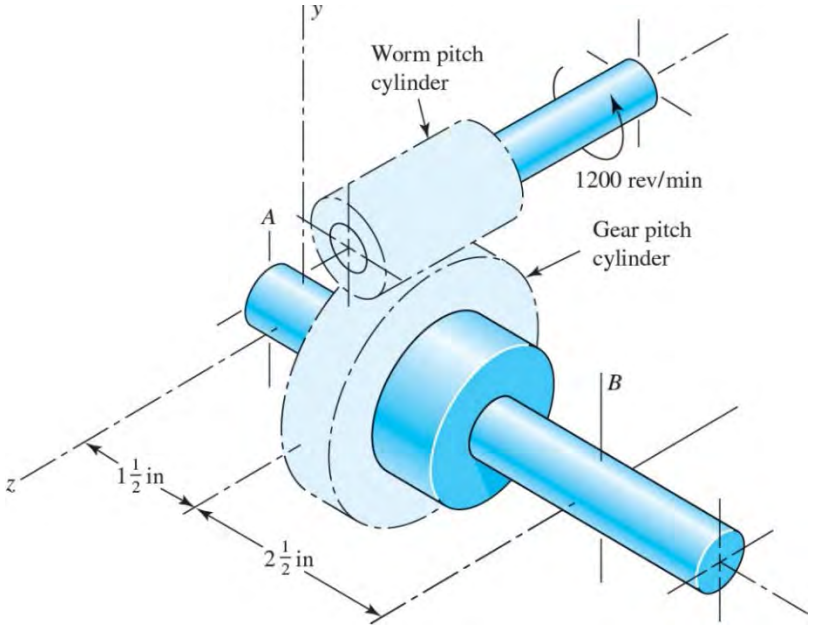
$$V_G = V_S sin\lambda$$

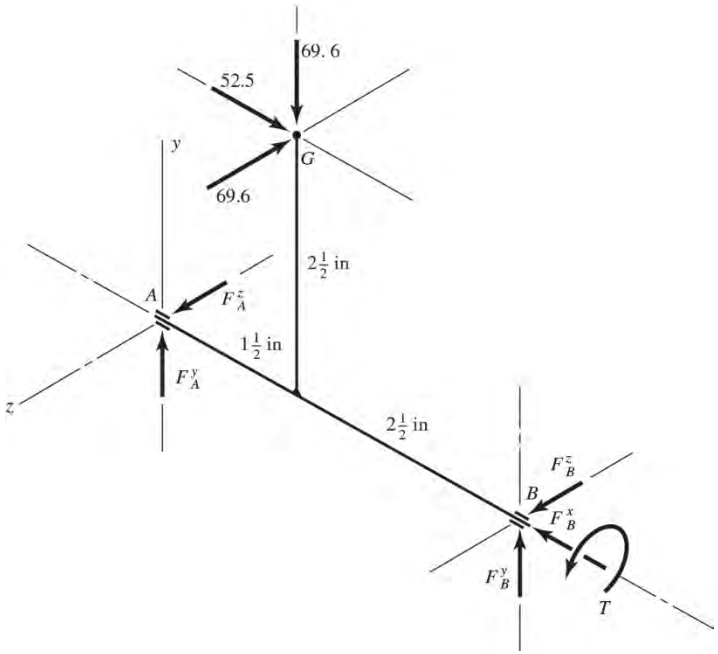


$$V_s = \frac{\pi n_W d_m}{12 \cos \lambda}$$

$$T_G = \frac{W^t D_m}{2}$$

Worm Gear Forces





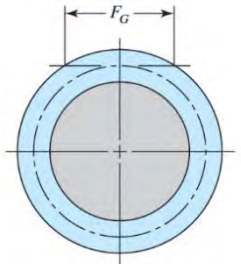
$$W^x = W(\cos\phi_n \sin\lambda + f\cos\lambda)$$
$$W^y = W\sin\phi_n$$
$$W^z = W(\cos\phi_n \cos\lambda - f\sin\lambda)$$

(13 - 43)

Worm Gear Durability

- AGMA gives a method for rating the surface durability of hardened steel worms operating on bronze gears.
- The method provides for a maximum allowable tangential force
- The ratings are based on minimum wear and pitting on the gear
- The gear will wear faster than the worm

$$\left(W^t\right)_{\text{all}} = C_s D_m^{0.8} F_e C_m C_v \qquad \textbf{(15 - 28)}$$



where C_s = materials factor

D_m = mean gear diameter, inches

F_e = effective face width of the gear (actual face width, F_G, but not to exceed $0.67 d_m$, the mean worm diameter), inches

C_m = ratio correction factor

C_v = velocity factor

- Compare $(W^t)_{all}$ with the W^t_{Gear} to ensure the required safety factor

Worm Gear Materials Factor, C_s

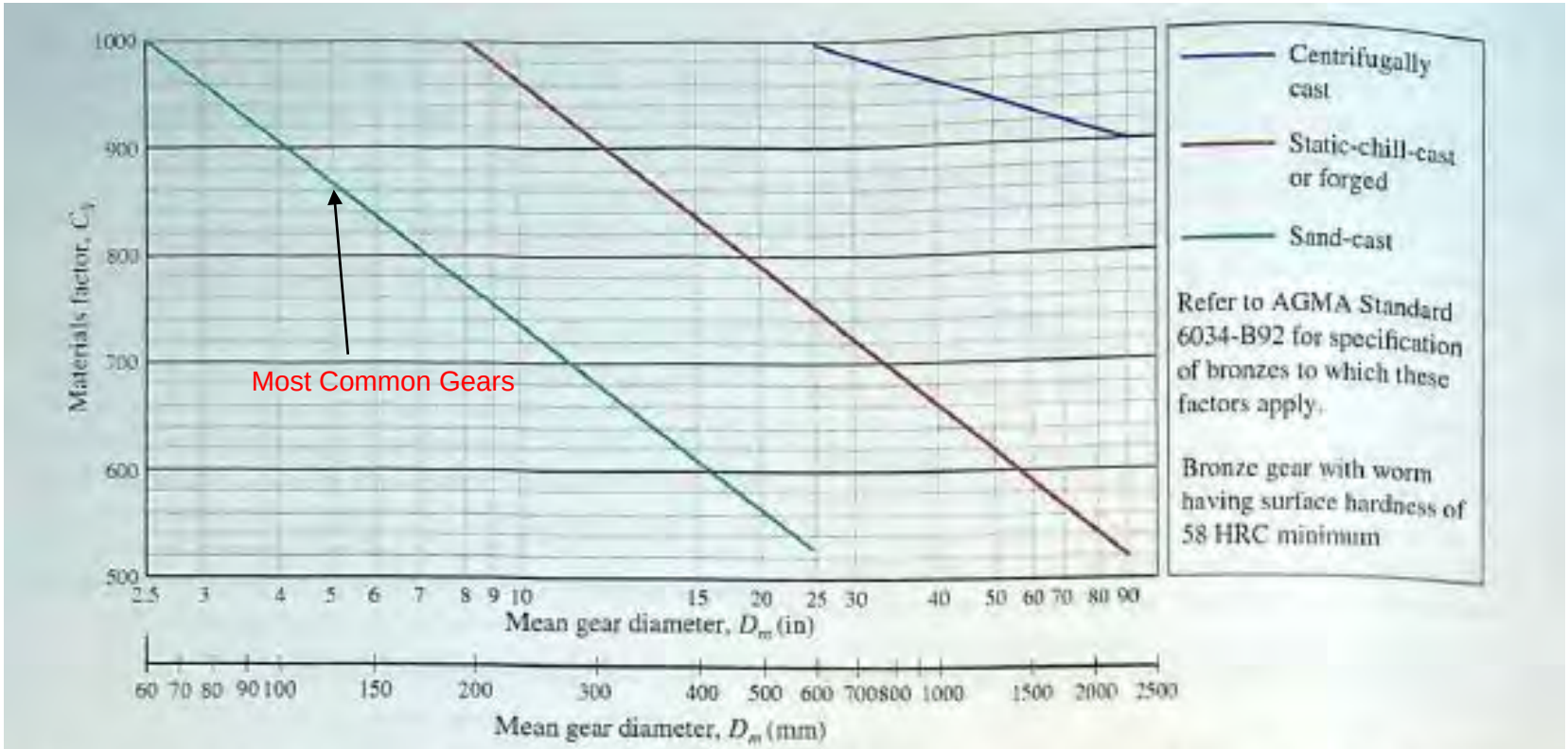


FIGURE 10–27 Materials factor, C_s, for hardened steel worms operating with bronze gears for center distance $C > 3.0$ in (76 mm) (Extracted from AGMA Standard 6034–B92, *Practice for Enclosed Cylindrical Wormgear Speed Reducers and Gearmotors*, with permission of the publisher, American Gear Manufacturers Association, 1001 North Fairfax Street, 5th Floor, Alexandria, VA 22314.)

Worm Gear Materials Factor, C_s

Worm Gearing Equations for Allowable Tangential Force [2]

$$C_s = 720 + 10.37C^3 \quad C \le 3 \text{ in} \qquad \textbf{(15-32)}$$

C = Center Distance

For sand-cast gears,

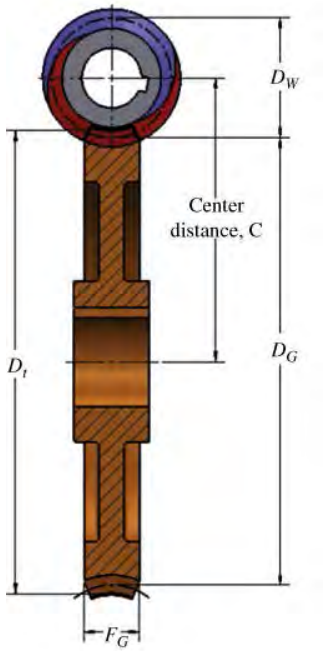
$$C_s = \begin{cases} 1000 & C > 3 \quad D_m \le 2.5 \text{ in} \\ 1190 - 477 \log D_m & C > 3 \quad D_m > 2.5 \text{ in} \end{cases} \qquad \textbf{(15-33)}$$

For chilled-cast gears,

$$C_s = \begin{cases} 1000 & C > 3 \quad D_m \le 8 \text{ in} \\ 1412 - 456 \log D_m & C > 3 \quad D_m > 8 \text{ in} \end{cases} \qquad \textbf{(15-34)}$$

For centrifugally cast gears,

$$C_s = \begin{cases} 1000 & C > 3 \quad D_m \le 25 \text{ in} \\ 1251 - 180 \log D_m & C > 3 \quad D_m > 25 \text{ in} \end{cases} \qquad \textbf{(15-35)}$$



D_m = Mean Gear Diameter ≈ D_G

- Shigley Equations 15-32 to 15-35

Worm Gear Ratio Correction Factor, C_m

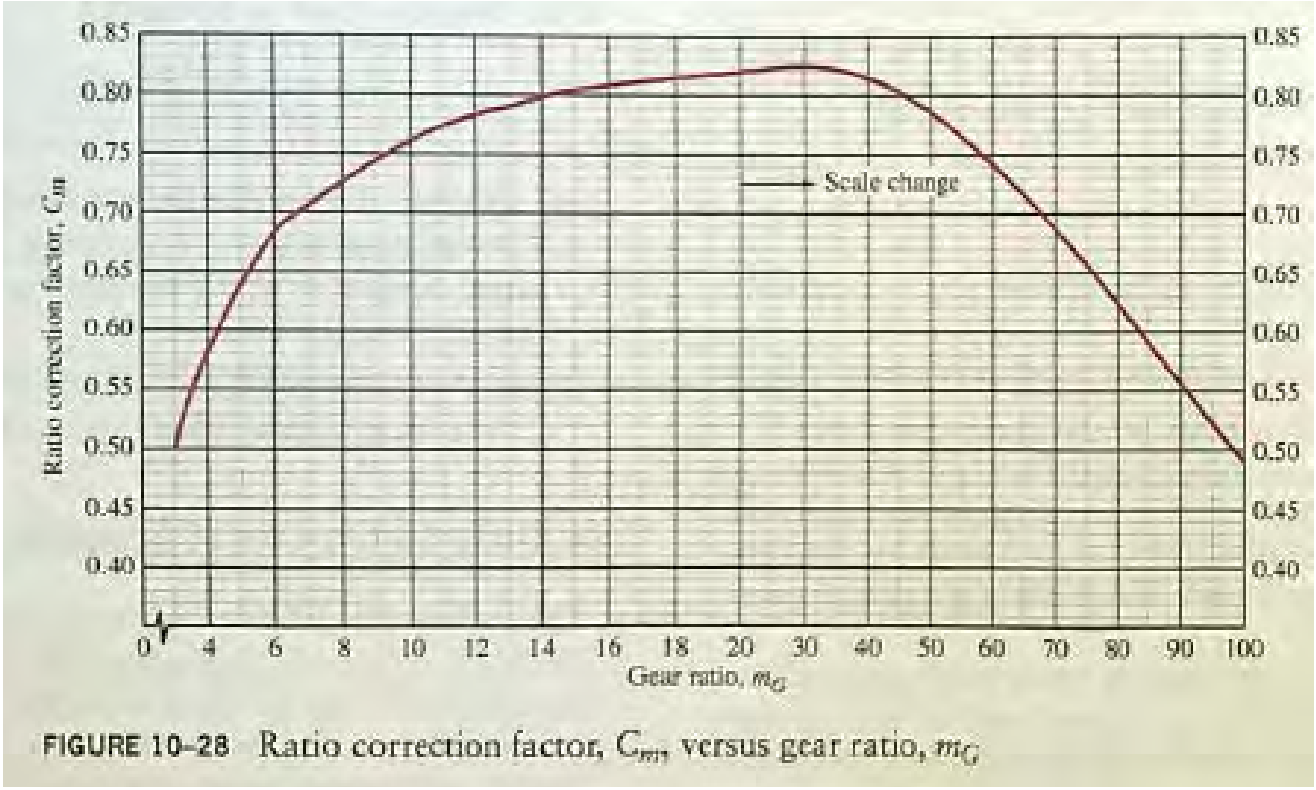


FIGURE 10–28 Ratio correction factor, C_m, versus gear ratio, m_G

$$C_m = \begin{cases} 0.02\sqrt{-m_G^2 + 40m_G - 76} + 0.46 & 3 < m_G \le 20 \\ 0.0107\sqrt{-m_G^2 + 56m_G + 5145} & 20 < m_G \le 76 \\ 1.1483 - 0.006\,58m_G & m_G > 76 \end{cases} \quad \textbf{(15-36)}$$

Worm Gear Velocity Factor, C_v

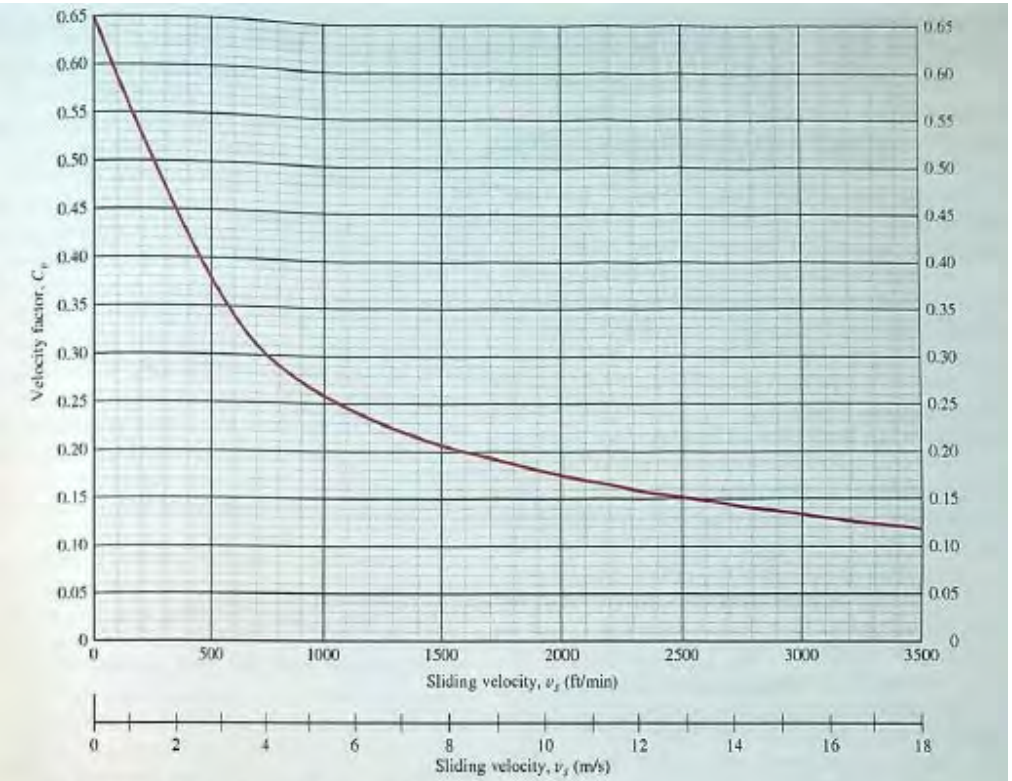


FIGURE 10–29 Velocity factor, C_v, versus sliding velocity

$$C_v = \begin{cases} 0.659\ \exp\left(-0.001\ 1 V_s\right) & V_s < 700 \text{ ft/min} \\ 13.31\ V_s^{-0.571} & 700 \le V_s < 3000 \text{ ft/min} \\ 65.52\ V_s^{-0.774} & V_s > 3000 \text{ ft/min} \end{cases} \qquad \textbf{(15-37)}$$

Bending Stress in Worm-gear Teeth

- Worm teeth are inherently much stronger than worm-gear teeth
- Stresses on worm-gears are only considered.
- Worm-gear teeth are short and thick on the edges of the face.
- Buckingham adapted the Lewis equation to determine the working stress,

$$\sigma_a = \frac{W_G^t}{p_n F_e y} \qquad \textbf{(15 - 53)}$$

- Y is the Lewis Form Factor. Y = 0.125 for $\phi = 20°$ (Table 10-5)
- Note: Mott uses a modified dynamic load value (See 10-11)
- The worm-gear stress is compared with the fatigue stress of the material
- For bronze gears use σ_{all} = 24ksi (phosphrous bronze –most common)

Heat Issues in Worm-gear Teeth

$$H_{\text{loss}} = 33\,000(1-e)H_{\text{in}} \qquad \textbf{(15 - 49)}$$

- Overall coefficient for combined convective and radiative heat transfer from the worm-gear case,

$$\hbar_{\text{CR}} = \begin{cases} \dfrac{n_W}{6494} + 0.13 & \text{no fan on worm shaft} \\ \dfrac{n_W}{3939} + 0.13 & \text{fan on worm shaft} \end{cases} \qquad \textbf{(15 - 50)}$$

- With case lateral area *A*, the oil sump temperature,

$$t_s = t_a + \frac{H_{\text{loss}}}{\hbar_{\text{CR}} A} = \frac{33\,000\left(1-e\right)\left(H\right)_{in}}{\hbar_{\text{CR}} A} + t_a \qquad \textbf{(15 - 51)}$$

- AGMA recommended minimum lateral area in in^2

$$A_{\text{min}} = 43.20C^{1.7} \qquad \textbf{(15 - 52)}$$